

Maschinelles Lernen für die Chemie - Exercise 1

Crash Courses: Python, Pandas and Numpy

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Introduction

Aim of the Exercise

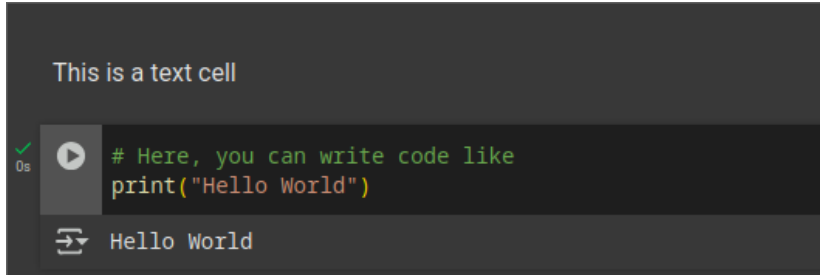
- Basic Python programming
- Solving small machine learning problems in Python

Structure

- Short presentation of relevant topics (10-15min)
- Afterwards: free time to program
- Ask questions, discuss with others, take breaks

Google Colab and Jupyter Notebooks

- We will use **Jupyter Notebooks**:



- Edit the notebooks in **Google Colab**: <https://colab.research.google.com/>, more information in `student_instructions.md` (ILIAS)

Overview: Exercises

- Notebooks are uploaded to ILIAS one week before the exercise
- Tutorial notebooks: read the text, click through the cells, try to understand the concepts and how the code works
- Exercise notebooks: apply the topics from the tutorial notebooks to a similar problem
- After the exercise: solution notebooks for the exercises are uploaded to ILIAS
- Questions: ask in the exercise, write in the ILIAS forum or mail at leonie.pfeffer@kit.edu or maike.vogelbacher@kit.edu

Dates

Every two weeks 12-3pm in the computer room (30.44, 4th floor):

- 28.04. (today)
- 12.05.
- 26.05.
- ~~09.06.~~ (holiday)
- 23.06.
- 07.07.
- 21.07.

Additional Resources

- bwGPT (alternative to ChatGPT): <https://bwgpt.scc.kit.edu/login?>
⇒ log in with your KIT account
- Various forums, just search for your problem in your favorite search engine
- Numpy / Pandas / Sklearn documentation

numpy.mean

```
numpy.mean(a, axis=None, dtype=None, out=None, keepdims=<no value>, *,  
where=<no value>) \[source\]
```

Compute the arithmetic mean along the specified axis.

Today's Topics

- Python basics (`0.1_Python_Crashkurs.ipynb`)
- Pandas tutorial and exercise (`1.1_Pandas_Crashcourse.ipynb` and `1.2_Pandas_Exercise.ipynb`)
- Numpy tutorial (`2.1_Numpy_Crashcourse.ipynb`)

Demonstration

Jupyter Notebook in Google Colab...

Summary: `student_instructions.md`